Thus, when we say $\lim_{x \to x_0} f(x)$ exists, it is understood that L is a unique real number. If any one of the above conditions fails then we say, the limit of $f(x)$ as x approaches x_0 does not exist.

We remark that the existence of one sided limits is weaker than the existence of limits.

Sometimes it is very useful to use the following in computing left and right limits. For $h > 0$,

$f(x_0^-) = \lim_{h \to 0} f(x_0 - h)$ and $f(x_0^+) = \lim_{h \to 0} f(x_0 + h)$.

Note that $f(x_0^-)$ and $f(x_0^+)$ stand for the left and right limiting values. But $f(x_0)$ is the value of the function at $x = x_0$.

Example 9.1

Calculate $\lim_{x \to 0} |x|$.

Solution

Recall from the earlier chapter 1

that $|x| = \begin{cases} -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ x & \text{if } x > 0 \end{cases}$

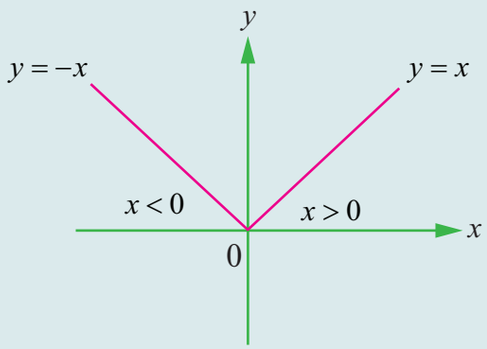


Fig. 9.10

If $x > 0$, then $|x| = x$, which tends to 0 as $x \to 0$ from the right of 0. That is, $\lim_{x \to 0^+} |x| = 0$

If $x < 0$, then $|x| = -x$ which again tends to 0 as $x \to 0$ from the left of 0. That is, $\lim_{x \to 0^-} |x| = 0$.

Thus, $\lim_{x \to 0^-} |x| = 0 = \lim_{x \to 0^+} |x|$.

Hence $\lim_{x \to 0} |x| = 0$.

Example 9.2

Consider the function $f(x) = \sqrt{x}$, $x \geq 0$.

Does $\lim_{x \to 0} f(x)$ exist?

Solution

No. $f(x) = \sqrt{x}$ is not even defined for $x < 0$.

Therefore as $x \to 0^-$, $\lim_{x \to 0^-} \sqrt{x}$ does not exist.

However, $\lim_{x \to 0^+} \sqrt{x} = 0$. Therefore $\lim_{x \to 0} \sqrt{x}$ does not exist.

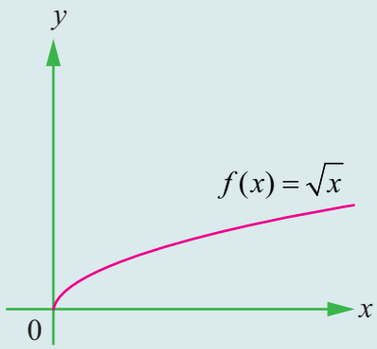


Fig. 9.11

Does $\lim_{x \to 0^-} \log x$ exist?

Look at the graph of $\log x$ for the answer.

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